

CSE Alpha Engine — System Architecture & Process Specification

Version: 1.0 **Scope:** Web application that runs Colombo Stock Exchange equities through four independent quantitative models, fuses the outputs, and presents a decision surface where the human makes the final call. **Design date:** August 2026

0. The One-Page Version

What the system does: ingests point-in-time CSE price, fundamental and macro data → runs four models that each answer a *different* question → fuses them into a single ranked, conviction-tiered list with entry bands, exit triggers and position sizes → shows the user the evidence behind every number so they can override it.

The four models are not four opinions on the same question. They have separate jobs:

Model	Question it answers	Analogy	Role
M1 — Carhart 4-factor	Is this return actually skill, or just paid-for factor exposure?	The referee	Filter + risk attribution
M2 — Ratio / value-quality screen	<i>What</i> is worth owning?	The scout	Cross-sectional selection
M3 — Momentum / contrarian	<i>When</i> do I buy and sell it?	The timer	Entry/exit timing
M4 — Macro regime (ARDL/VECM/VAR)	<i>How much</i> should I own, and which style?	The weather forecast	Exposure + style rotation

A scout with no weather forecast buys great companies into a bear market. A timer with no scout chases junk. A referee with no scout has nothing to judge. **The architecture only works if each model stays in its lane.**

The binding constraint — read this before anything else. CSE all-in transaction costs are roughly <cite index="2-1">1.12% one way for trades up to LKR 100m (0.640% brokerage +

0.300% share transaction levy + 0.084% CSE + 0.072% SEC cess + 0.024% CDS), i.e. approximately 2.24% for a round trip</cite>. Add bid-ask spread and market impact in a thin book and your realistic all-in friction is **3–5% per round trip on mid-caps**. That is your hurdle rate before you have made a single rupee. Every architectural decision below is downstream of this one fact: *this must be a low-turnover, high-conviction, medium-horizon system*. Any design that trades weekly is dead on arrival regardless of model quality.

Target metrics (the system is only “working” if it hits these):

Metric	Target	Kill threshold
Hit rate, 6-month horizon	≥ 58%	< 50% over 20 closed positions
Avg win / avg loss	≥ 1.8×	< 1.2×
Net annual return vs ASPI	+10–15pp	Underperform 2 consecutive years
Annual turnover	≤ 150%	> 250% (costs eat you)
Max drawdown	≤ 25%	> 35%
Live IR ÷ backtest IR	≥ 0.5	< 0.3 over 12m

1. Edge Thesis — Why This Can Actually Work on the CSE

You do not get an edge by running a Carhart regression. Institutions everywhere do that. You get an edge on the CSE because of **structural features of this specific market** that the models are configured to exploit:

CSE structural feature	Exploitable consequence	Which model harvests it
~280 listed companies, thin analyst coverage below the SL20	Fundamental mispricing persists for months, not days	M2
Retail/sentiment-dominated flow, low institutional arbitrage	Strong short-term overreaction <i>and</i> slow medium-term information diffusion	M3
Extreme sensitivity to T-bill yields — equity is treated as a fixed-income substitute	Macro turning points produce market-wide re-ratings that are forecastable with a 1–3 month lead	M4

Family-controlled conglomerates, opaque related-party flows	Earnings-quality screening is unusually high-value; avoiding the bad half beats picking the good half	M2
Illiquidity	Deters institutional capital → the mispricing is <i>not</i> arbitrated away. This is the moat.	Universe gates
No commercial factor library for Sri Lanka	Building CSE-native SMB/HML/MOM factors is itself an edge	M1 / Factor layer

The honest version: your edge is *capacity-constrained*. It works at LKR 5m–200m. It degrades badly above that. That is fine — it means large funds structurally cannot compete with you here. Illiquidity is the fence around your field, not a bug.

2. System Architecture — Layer Map

L8	DECISION SURFACE (React)
	Regime dashboard · Screener · Stock deep-dive · Portfolio & risk · Backtest lab · Journal
L7	MONITORING — drift, decay, kill switches, alerting
L6	EXECUTION LOGIC — entry bands, exits, sizing, costs
L5	SIGNAL FUSION — sequential gating → composite score
L4	MODEL ENGINES
	M1 Carhart M2 Ratio screen M3 Mom/Rev M4 Macro
L3	FACTOR LIBRARY — CSE-native MKT, SMB, HML, MOM, LIQ
L2	UNIVERSE CONSTRUCTION — liquidity & integrity gates
L1	DATA — point-in-time store (prices, fundamentals, macro, corporate actions, events, disclosures)

Data flows up. Constraints flow down. **L4 never sees a stock that L2 rejected. L6 never sizes a position larger than L2's liquidity cap allows.**

3. Layer 1 — Data

3.1 Sources

Domain	Source	Frequency	Method
Prices, volume, turnover, trades	CSE website / broker feeds	Daily EOD	Apify actor → normalised loader
Company financials (quarterly + annual)	CSE company disclosures (PDF/XBRL)	Quarterly	Scrape + PDF table extraction + manual QA queue
Corporate actions (splits, rights, bonus, dividends)	CSE announcements	Event-driven	Scrape + mandatory human confirm
Shareholding / free float / top-20 holders	Annual reports, quarterly disclosures	Quarterly	Scrape
Foreign net flow	CSE daily market summary	Daily	Scrape
Policy rate, AWPLR, T-bill yields, M2b, credit growth, reserves	CBSL	Daily / weekly / monthly	API + scrape
CCPI headline & core, NCPI	DCS / CBSL	Monthly	Scrape
LKR/USD, LKR/EUR, NEER/REER	CBSL	Daily	Scrape
Tourist arrivals, remittances, trade balance	CBSL / SLTDA	Monthly	Scrape
Sovereign spread, ratings actions, IMF review milestones	IMF, rating agencies, news	Event-driven	Scrape + manual event log
Brent, global EM risk proxies	Public feeds	Daily	API

3.2 The non-negotiable rule: point-in-time (PIT)

Every fundamental record must be stamped with **three dates**:

1. `period_end` — the fiscal period the number describes
2. `first_available_date` — the date the market could first see it (release date)
3. `ingest_date` — when your system stored it

All models query on `first_available_date ≤ t`, never `period_end ≤ t`. Financial statements are also **immutable and versioned** — when a company restates, you insert a new row, you never update the old one. Restated data in a backtest is the single most common source of fake alpha.

CSE-specific timing: a large share of CSE companies use a **31 March fiscal year end**, and interim/annual filings can lag 45–90 days. Your default assumption should be a **conservative 3-month reporting lag on quarterlies and 6-month on audited annuals**, verified per company. Never let a March-year-end company's numbers enter a portfolio formed before end-September.

3.3 Schema (core tables)

<code>securities</code>	(<code>ticker</code> , <code>isin</code> , <code>name</code> , <code>gics_sector</code> , <code>listing_date</code> , <code>delisting_date</code> , <code>board</code> , <code>fiscal_year_end</code>)
<code>prices_daily</code>	(<code>ticker</code> , <code>date</code> , <code>open</code> , <code>high</code> , <code>low</code> , <code>close</code> , <code>vwap</code> , <code>volume</code> , <code>turnover</code> , <code>trades</code> , <code>adj_factor</code>)
<code>corporate_actions</code>	(<code>ticker</code> , <code>ex_date</code> , <code>type</code> , <code>ratio</code> , <code>cash_amount</code> , <code>source_url</code>)
<code>fundamentals</code>	(<code>ticker</code> , <code>period_end</code> , <code>period_type</code> , <code>first_available_date</code> , <code>version</code> , <code>statement_line</code> , <code>value</code> , <code>currency</code> , <code>restated_flag</code>)
<code>float_data</code>	(<code>ticker</code> , <code>as_of</code> , <code>shares_issued</code> , <code>public_float_pct</code> , <code>top20_pct</code> , <code>controlling_holder</code>)
<code>macro_series</code>	(<code>series_id</code> , <code>obs_date</code> , <code>first_available_date</code> , <code>value</code>)
<code>events</code>	(<code>event_id</code> , <code>date</code> , <code>type</code> , <code>description</code> , <code>affected_scope</code>)
<code>factor_returns</code>	(<code>date</code> , <code>mkt_rf</code> , <code>smb</code> , <code>hml</code> , <code>mom</code> , <code>liq</code> , <code>rf</code>)
<code>model_outputs</code>	(<code>ticker</code> , <code>as_of</code> , <code>model_id</code> , <code>score</code> , <code>subscores_json</code> , <code>diagnostics_json</code> , <code>model_version</code>)
<code>positions / trades / journal</code>	

3.4 Corporate action integrity

The dividend/bonus/rights history is where CSE backtests silently die. A single unadjusted 1:1 bonus issue creates a fake -50% return that will look like a momentum signal.

- Build a **total-return adjustment factor** series per ticker, applied cumulatively backwards.
- Rights issues need theoretical ex-rights price (TERP) adjustment, not simple dividend adjustment.

- **Reconciliation test, run nightly:** recompute each stock's total return from adjusted prices vs. from raw prices + declared actions. Any mismatch > 0.5% raises a data alert and quarantines the ticker from all models until resolved.

3.5 Survivorship

Delisted, suspended, and defaulted companies **stay in the database and stay in every backtest** until their delisting date, with a terminal return (recovery value or -100%). CSE has had meaningful delisting activity; excluding it will overstate returns by several points a year.

4. Layer 2 — Universe Construction & Gates

Run daily. Produces the `investable_universe` that every model downstream operates on.

Gate 1 — Liquidity (hard)

Test	Threshold	Rationale
Median daily turnover, 60d	$\geq$ LKR 2.0m	Below this you cannot enter or exit at a rational price
Days traded, last 60 sessions	≥ 45 (75%)	Non-trading breaks every return calculation
Amihud illiquidity, 60d	Below universe 80th percentile	Direct impact-cost proxy
Your max position $\div$ 20d ADV	$\leq 15\%$	Defines the position cap, not just eligibility

Set thresholds as parameters, not constants — they should scale with your capital.

Gate 2 — Structural (hard)

- Free float $\geq 15\%$ of shares issued
- Not on Watch List / trading suspension
- Listed ≥ 12 months (no IPO pop chasing)
- Market cap $\geq$ LKR 1.0bn
- At least 8 quarters of continuous financial history

Gate 3 — Integrity (hard veto, applied later at M2 but flagged here)

- Qualified audit opinion or emphasis-of-matter on going concern → **exclude**
- Auditor change + CFO departure within same 12 months → **exclude**
- Beneish M-Score > -1.78 → **exclude**
- Related-party revenue or receivables > 30% of total → **exclude**

The universe should end up around **90–140 names** out of ~280 listed. That is normal and correct. A wide universe on the CSE is a liability, not an asset.

5. Layer 3 — CSE-Native Factor Library

No Fama-French library exists for Sri Lanka. You are building one. This is a genuine, durable asset — treat it as a product, not a preprocessing step.

5.1 Construction spec

Risk-free rate (Rf): 91-day Treasury bill weekly primary market yield, converted to the return frequency.

Market factor (MKT-RF): free-float value-weighted total return of the *investable universe*, minus Rf. **Do not use ASPI** — it is full market-cap weighted and dominated by non-tradable controlling stakes, so it does not represent an achievable portfolio.

Size and value (SMB, HML) — 2×3 Fama-French sort:

- **Formation date: 30 September each year** (not June). This gives a 6-month lag after the dominant 31 March fiscal year end, guaranteeing no look-ahead.
- Size split at universe **median** free-float market cap.
- Book-to-market split at **30th / 70th percentiles**, using book equity from the most recent audited annual report.
- **CSE adjustment — “hard book”:** strip revaluation reserves on land and buildings from book equity for hotel, plantation and property companies, and compute a second B/M on hard book. Property-revaluation-inflated book value is the #1 source of fake “value” on the CSE. Carry both HML and HML_hard; use HML_hard as the primary.
- $SMB = \frac{1}{3}(SmallValue + SmallNeutral + SmallGrowth) - \frac{1}{3}(BigValue + BigNeutral + BigGrowth)$
- $HML = \frac{1}{2}(SmallValue + BigValue) - \frac{1}{2}(SmallGrowth + BigGrowth)$

Momentum (MOM/UMD): 2×3 sort on size × prior 12-month return skipping the most recent month (t−12 to t−2). Rebalanced monthly.

Liquidity factor (LIQ) — add this, it is not optional here: long the least-liquid tercile, short the most-liquid tercile by Amihud, within size groups. On an illiquid frontier market the liquidity premium is often larger than HML, and if you omit it, it will masquerade as alpha in M1.

5.2 The non-synchronous trading correction

CSE stocks routinely go days without a trade. Standard OLS betas on such stocks are **severely downward biased**. Every beta in M1 must use the **Dimson (1979) aggregated-coefficient method**:

$$R_{i,t} - R_{f,t} = \alpha + \sum_{k=-1..+1} \beta_k \cdot \text{MKT}_{t+k} + [\text{same for SMB, HML, MOM, LIQ}] + \varepsilon$$
$$\beta_{\text{true}} = \beta_{-1} + \beta_0 + \beta_{+1}$$

Skipping this step is the single most common technical error in frontier-market factor work, and it will make illiquid small caps look like low-beta free lunches.

5.3 Estimation windows

Sri Lanka has a structural break in 2022 (sovereign default) that will contaminate any long window.

- **Primary:** weekly returns, 156-week (3-year) rolling window, Newey-West standard errors (lag 4).
- **Secondary:** monthly returns, 60-month window, reported alongside for stability comparison.
- Any regression spanning March 2022 – December 2023 must carry a **structural break flag** in the UI. Do not silently pool pre- and post-default data.

6. Layer 4 — The Four Model Engines

M1 — Carhart Four-Factor Model (*the referee*)

Job: decompose a stock's historical return into paid-for factor exposure vs. genuine unexplained return, and produce a forward expected return conditional on M4's factor-premium forecast.

Specification (5-factor in practice, Carhart + LIQ):

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{MKT}(MKT - Rf)_t + \beta_{SMB} \cdot SMB_t + \beta_{HML} \cdot HML_t + \beta_{MOM} \cdot MOM_t + \beta_{LIQ} \cdot LIQ_t + \varepsilon_{i,t}$$

with Dimson lead/lag aggregation and Newey-West errors.

Outputs per stock:

Output	Use
α (annualised) + t-stat	The alpha claim. Require t > 1.5 to count as evidence, not just a positive point estimate.
β_{MKT} , β_{SMB} , β_{HML} , β_{MOM} , β_{LIQ}	Style fingerprint. Feeds portfolio exposure control.
R^2	Below 0.15 → the model explains almost nothing; treat α as noise, downweight M1 for this name.
Residual (idiosyncratic) volatility	Primary input to position sizing.
Rolling 36m α path	Alpha stability. A decaying α is a sell signal, a spiky α is data error.
Expected return = $R_f + \sum \beta_k \cdot E[\text{premium}_k \text{regime}]$	Forward-looking return estimate from M4's regime.

Critical interpretation rule — this is where most people misuse Carhart. A stock with high raw returns and $\beta_{MOM} = 0.9$ is **not a good pick, it is a momentum bet you are already making elsewhere**. M1's job is to stop you double-counting. Specifically:

- If M3 has flagged a stock on momentum **and** M1 shows high β_{MOM} with near-zero α → the signal is *redundant*, not confirmed. Downgrade it.
- If M2 has flagged a stock on value **and** M1 shows high β_{HML} with near-zero α → you are buying the value factor, not a special company. Acceptable, but size it as a factor bet, not a conviction pick.
- **The prize case:** M2 or M3 flags it, and M1 shows **positive α with modest loadings**. That is genuine idiosyncratic mispricing. These are your highest-conviction names.

Portfolio-level use: run the same regression on your live portfolio's return stream. If aggregate $\beta_{HML} > 1.2$ or $\beta_{MOM} > 0.8$, you have accidentally become a single-factor fund. Trigger a diversification alert.

M2 — Financial-Ratio Predictive / Value-Quality Screen (*the scout*)

Job: rank the universe on the probability of positive 6–12 month excess return, using fundamentals only.

Four pillars, each a cross-sectional percentile z-score, winsorised at 1%/99%, computed *within sector*:

Pillar A — Value (30%)

- EV / EBIT
- EV / EBITDA
- P / Hard Book (revaluation-adjusted)
- Free cash flow yield (FCF / EV)
- Earnings yield (normalised, 3-year average EBIT to smooth cyclicality)
- **Sum-of-the-parts NAV gap** — for conglomerates (JKH, Hayleys, Melstacorp, Aitken Spence, etc.), value listed subsidiaries at market, unlisted at sector multiples, subtract net holdco debt, apply a holding-company discount. This is *the* CSE-native value metric and deserves its own weight.

Pillar B — Quality (30%)

- ROIC and ROIC – WACC spread
- Gross profitability (gross profit ÷ total assets) — Novy-Marx; robust in noisy-earnings markets
- 3-year ROIC stability (inverse of standard deviation)
- Cash conversion: CFO ÷ net income (3-year average)
- **Piotroski F-Score** (0–9) — carried as its own displayed number, high signal on cheap small caps

Pillar C — Safety (20%)

- Net debt / EBITDA
- Interest coverage (EBIT / interest expense)
- Altman Z"-Score (emerging-market variant)
- Refinancing cliff: short-term debt ÷ (cash + CFO)
- FX exposure: unhedged foreign-currency debt ÷ total debt — **critical in Sri Lanka**; a

10% LKR depreciation has repeatedly destroyed the equity of importers and USD-borrowers

Pillar D — Earnings Integrity (20%, and a veto)

- Beneish M-Score
- Sloan accrual ratio: $(\text{net income} - \text{CFO}) \div \text{average total assets}$
- Related-party revenue and receivables as % of total
- Days sales outstanding trend (3-year)
- Auditor tier and any opinion modification
- Director/insider net buying over 12 months

Two model variants, run in parallel — always:

1. **Baseline (equal-weight composite z-score).** Transparent, robust, and hard to beat. This is your production default.
2. **Learned model.** Elastic Net or heavily-regularised gradient boosting predicting forward 6-month sector-relative excess return, trained on purged walk-forward folds.

The learned model only replaces the baseline in production if it beats it out-of-sample on rank IC across at least three separate walk-forward folds. With ~130 names and ~12 years of usable history you have on the order of a few thousand observations — that is small-data territory, and unregularised ML will overfit spectacularly. Discipline here is worth more than model sophistication.

Banks and finance companies need a separate model. EV/EBIT and ROIC are meaningless for them. Substitute pillar A/B with: P/TBV vs ROE regression residual, NIM trend, CASA ratio, Stage 3 loans and provision cover, cost-to-income, Tier 1 CAR headroom, credit cost normalisation. Route by GICS sector at the model-selection step. Given how large banks are in the SL20, do not skip this.

Output: `m2_score` (0–100 percentile), four pillar sub-scores, F-Score, SOTP gap %, and an explicit **red-flag list** rendered in the UI.

M3 — Momentum / Contrarian Engine (*the timer*)

Job: answer “is now the moment?” — and, separately, “has the moment passed?”

The central design principle: horizon separation. Momentum and reversal are not competing theories; they operate at *different horizons*. The engine runs both and never lets

them fight.

Horizon	Effect	Signal
1 month	Reversal (overreaction)	Prior 1-month return, reversed
3–12 months	Momentum (slow diffusion)	12–2 and 6–1 cumulative return
36–60 months	Reversal (long-run over-extrapolation)	Prior 3–5 year return, reversed

Signal battery:

Signal	Definition	Weight
MOM_12_2	Cumulative return t–12 to t–2	20%
MOM_6_1	Cumulative return t–6 to t–1	15%
REV_1M	$-1 \times$ prior month return	15%
52-week high proximity	Price $\div$ 52-week high (George–Hwang)	20%
Residual momentum	12–2 momentum computed on M1's residuals ϵ , not raw returns	20%
Volume confirmation	20d turnover $\div$ 120d turnover, z-scored	10%

Two notes on the highest-weighted items. **52-week high proximity typically outperforms raw momentum in frontier markets** — anchoring is a strong behavioural force in retail-dominated books, and the metric is far more robust to the stale-price problem than cumulative return. **Residual momentum** strips out the factor component, so what remains is stock-specific drift — exactly the thing M1 is trying to certify as alpha. These two are the core of M3; the rest are supporting evidence.

Contrarian branch — activation conditions (all must hold):

1. REV_1M in the bottom decile (a sharp recent selloff), **and**
2. M2 score ≥ 70 (fundamentals intact), **and**
3. No red flags in M2 Pillar D, **and**
4. No negative earnings revision or adverse disclosure in the last 60 days, **and**
5. Selloff not attributable to a sector-wide macro shock still in force per M4

This is the “**baby thrown out with the bathwater**” trade. On the CSE it is high-hit-rate precisely because retail selling is indiscriminate — but only when the fundamentals genuinely are intact. Conditions 3 and 4 are what stop it from becoming a falling-knife machine.

Momentum crash guard. Momentum strategies suffer catastrophic drawdowns during sharp rebounds off market bottoms in high-volatility states. Hard rule: **when M4 reports regime = Risk-Off with rising transition probability to Risk-On, momentum weight is cut to 25% and contrarian weight is raised.** M3 subscribes to M4’s regime output — it is not independent of it.

Output: `M3_score` (0–100), branch label (`momentum` / `contrarian` / `neutral`), the signal battery breakdown, and a **timing verdict:** `enter now` / `wait for pullback` / `wait for confirmation` / `exit` .

M4 — Macro-Economic Regime Model (*the weather forecast*)

Job: set gross exposure, style tilt and sector tilt. This layer is worth more than the other three combined on the CSE, because Sri Lankan equities move as a *bloc* with the rate cycle far more than they move on idiosyncratic news.

The core Sri Lankan macro insight, stated plainly: CSE equity is priced as a **substitute for T-bills**. When the 364-day yield falls, retail and institutional money rotates into equities en masse and the market re-rates regardless of earnings. The **equity earnings yield minus 364-day T-bill yield spread** is therefore your single most powerful macro variable. Build the whole model around it. As of mid-2026 the relevant backdrop is a policy rate held at [8.75%](#), [GDP growth of 5% in 2025](#), inflation returned to positive territory and reserves around [US\\$7bn](#), and [an IMF EFF program with a combined fifth and sixth review reached at staff level in April 2026](#) — a *reflation-with-external-risk* configuration.

Variable set:

Block	Series
Monetary	Overnight Policy Rate, AWPLR, 91d and 364d T-bill yields, M2b growth, private sector credit growth
Prices	CCPI headline y/y, CCPI core y/y, NCPI
External	LKR/USD, REER, gross official reserves, trade balance, tourist arrivals, remittances, Brent

Fiscal / sovereign	Primary balance, sovereign spread / rating actions, IMF review milestones
Market internals	ASPI turnover, foreign net flow, advance-decline, market-wide earnings yield spread

Method chain (execute in this order):

Step 1 — Stationarity and breaks. ADF + Phillips-Perron + KPSS on every series. Then **Zivot-Andrews** with an endogenous break — Sri Lanka's 2022 default is a structural break in nearly every series, and ignoring it will make you conclude "no cointegration" when a relationship exists in both sub-periods.

Step 2 — Choose the estimator by the integration profile.

Condition	Estimator
All variables $I(1)$ and Johansen finds cointegration	VECM — gives long-run equilibrium + error-correction speed
Mixed $I(0)/I(1)$, none $I(2)$, short sample	ARDL bounds test (Pesaran–Shin–Smith) ← <i>this will be your default</i>
No cointegration found	VAR in first differences

ARDL is the correct default here for three reasons: Sri Lankan macro series are a genuine mix of $I(0)$ and $I(1)$; your usable post-liberalisation sample is short; and ARDL handles small samples better than Johansen. It also gives you the two things you actually want in one estimation: a **long-run cointegrating relationship** (where should the market be, given the macro state?) and an **error-correction term** (how fast does it get there?). An ECT coefficient of, say, -0.28 with monthly data means ~28% of the gap closes per month — a half-life of about 2.1 months, which is directly actionable as a holding period.

Step 3 — Dynamics. Impulse response functions (12–24 month horizon, Cholesky ordering justified by economic exogeneity: global → policy → domestic real → financial → market), forecast error variance decomposition, and **Toda–Yamamoto Granger causality** (which is valid regardless of integration order — use it instead of standard Granger tests).

Step 4 — Regime classification. Two-state or three-state **Markov-switching model** on market returns and volatility, augmented with a macro composite z-score. States:

Regime	Signature	Gross exposure	Style tilt

Risk-On (reflation)	Falling yields, stable LKR, positive credit growth, earnings yield spread wide	85–100%	Momentum + small cap + high beta
Transition	Mixed signals, rising uncertainty	50–70%	Quality + large cap
Risk-Off	Rising yields, LKR depreciation, reserve drain, spread compression	20–40%	Quality + low leverage + defensive sectors + cash

Report the **regime probability**, not just the label. A 55/45 split is a fundamentally different instruction from a 90/10 split, and the UI must show that.

Step 5 — Event study module. For each recurring event class, estimate cumulative abnormal returns using a –250 to –30 estimation window and a **–5 to +20 event window** (wider than developed-market convention, because information diffuses slowly on the CSE and the post-event drift is where the tradable edge sits).

Event classes: CBSL policy announcements, CCPI releases, IMF review outcomes, budget presentations, elections, sovereign rating actions, tax policy changes (VAT, capital gains, import duty), major FX moves (>3% in a month).

Step 6 — Sector sensitivity matrix. Regress each sector’s excess return on macro shocks (rate surprise, FX shock, inflation surprise, oil shock) to produce a live table of *which sectors gain and lose from each shock*. This is the practical output that drives sector tilt.

Illustrative structure — populate from your own estimation, do not hard-code:

Sector	Rate cut	LKR depreciation	Oil spike	Tourism rebound
Banks	+ (re-rating) / – (NIM)	–	~	+
Diversified holdings	++	mixed	–	+
Hotels	+	+ (USD earners)	–	++
Manufacturing (import input)	+	--	--	~
Exporters (apparel, tea)	+	++	–	~
Construction / materials	++	–	–	~

Telecom	+	– (USD capex)	~	+
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Outputs: regime label + probability vector, recommended gross exposure %, style weight vector, sector tilt vector, 3/6/12-month directional forecast with confidence interval, ECT half-life, upcoming event calendar with historical CAR patterns.

7. Layer 5 — Signal Fusion

Do not average the four scores. They are not comparable quantities. Fusion is **sequential gating**, and the order matters.

STAGE 1 – M4 sets the frame

- gross_exposure (0-100%)
- style_weights {value, momentum, quality}
- sector_tilts
- If regime = Risk-Off with $P > 0.7$: cap new entries, raise quality weight, no new small-cap positions.

STAGE 2 – M2 builds the longlist

- Keep $M2_score \geq 65$ (sector-relative)
- Hard veto on any Pillar D red flag, regardless of other scores
- Apply M4 sector tilts to the ranking
- Result: 20-35 names

STAGE 3 – M3 times it

- For each longlist name, produce timing verdict
- 'wait' names go to a watchlist with a trigger price, not the buy list
- Result: 8-15 actionable names

STAGE 4 – M1 certifies and sizes the risk

- Reject names where the entire signal is explained by factor loading (α t-stat < 0.5 AND the dominant β aligns with the flagging model)
- Compute expected return under M4's regime-conditional premia
- Supply residual volatility for position sizing
- Result: 5-10 ranked, conviction-tiered candidates

Composite score (for ranking within the final set):

$$\begin{aligned}
 \text{Composite} = & 0.40 \cdot M2_score \\
 & + 0.30 \cdot M3_score \\
 & + 0.20 \cdot M1_alpha_percentile \\
 & + 0.10 \cdot M4_sector_fit_score
 \end{aligned}$$

with style-conditional weight adjustment:

Risk-On regime: M3 weight $\times 1.3$, M2 weight $\times 0.85$

Risk-Off regime: M2 weight $\times 1.3$ (quality pillar $\times 1.5$), M3 weight $\times 0.6$

Conviction tiers and the profitability hurdle:

Tier	Requirement	Target position
Tier 1	Composite ≥ 80 , M1 α t-stat > 1.5 , all four models aligned, expected 6m return $\geq 15\%$	8–10% of portfolio
Tier 2	Composite 70–80, three of four aligned, expected 6m return $\geq 10\%$	4–6%
Tier 3	Composite 60–70, expected 6m return $\geq 8\%$	2–3%
Reject	Expected 6m return $< 8\%$	—

That 8% floor is not arbitrary. With ~3–5% realistic all-in round-trip friction on a mid-cap, an 8% expected gross return is roughly a 4% net return for two positions' worth of risk.

Anything below that floor is a losing trade even if the model is right. This single rule will do more for your profitability than any refinement to the models.

8. Layer 6 — Entry, Exit, Sizing

Entry

- **Limit orders only.** Never market orders on the CSE — the book is too thin and you will pay 2–3% in a single fill.
- Entry band: composite fair value estimate minus required margin of safety (20% Tier 1, 25% Tier 2, 30% Tier 3).
- **Participation cap: $\leq 15\%$ of 20-day ADV per session.** A full position may take 3–10 sessions to build. The system must model this and show `estimated_days_to_build`.
- Staged entry: 40% / 35% / 25% across the accumulation window, with the second and third tranches conditional on the thesis remaining intact.

Exit — five independent triggers, any one fires

--	--

Trigger	Rule
Valuation	Price reaches fair-value ceiling (SOTP, or sector multiple reversion)
Thesis break	Any new M2 Pillar D red flag, or M2 score drops below 45 → exit regardless of P&L
Momentum break	M3 verdict flips to <code>exit</code> , or price closes below 100-day MA on above-average volume
Macro regime flip	M4 shifts to Risk-Off with $P > 0.7$ → reduce all positions to 50% of target
Time stop	12 months with no thesis progression → exit and redeploy. Capital has an opportunity cost.

No fixed percentage stop-loss. On an illiquid market with 5%+ daily swings on tiny volume, a fixed stop is a guaranteed donation to whoever is watching the book. Use thesis-based and structural exits.

Position sizing — take the minimum of four caps

```
size = MIN(
    risk_parity      = (target_portfolio_vol × risk_budget) ÷ M1_residual_vol,
    liquidity_cap    = (0.15 × ADV_20d × acceptable_exit_days) ÷ portfolio_value,
    concentration_cap= tier_max (10% / 6% / 3%),
    sector_cap       = 30% sector - existing sector exposure
)
```

The liquidity cap will bind more often than you expect, and **that is the system working correctly**. A model screaming Tier 1 on a stock that trades LKR 800k a day is telling you about a great company you cannot own at size.

9. Layer 7 — Backtest & Validation Protocol

You will generate false alpha unless every one of these is enforced. This section is the difference between a system that works and a system that looks like it works.

1. **Point-in-time only.** Every query filters on `first_available_date` . Restated financials never enter.
2. **Survivorship-inclusive.** Delisted and suspended names carry terminal returns.

3. **Purged walk-forward cross-validation.** Train on window W , purge the following 6 months (the forward-return horizon) to prevent label leakage, then test. Never random k -fold on time series.
 4. **Realistic cost model, applied always.** 1.12% each way in fees, **plus** a slippage term scaled to Amihud illiquidity and your order size relative to ADV. Report gross and net side by side, and treat the net number as the only real one.
 5. **Deflated Sharpe Ratio (Bailey & López de Prado).** You are testing dozens of signal variants. Report the DSR that corrects for the number of trials. If $DSR \leq 0$, you have found nothing.
 6. **Regime coverage.** The backtest must span at least three distinct macro regimes, and performance must be reported *per regime*, not just in aggregate. A strategy that only works in reflation is a leveraged bet on rate cuts.
 7. **Capacity curve.** Re-run the backtest at LKR 5m / 25m / 100m / 500m of capital. Plot alpha decay against AUM. This tells you the size at which your edge dies.
 8. **Sensitivity sweep.** Vary every key parameter (thresholds, windows, weights) $\pm 30\%$. If performance collapses, you have curve-fitted. A robust strategy is a *plateau*, not a peak.
 9. **Baseline comparisons — mandatory.** Report against: ASPI buy-and-hold, S&P SL20 buy-and-hold, equal-weight investable universe, and the simple M2 baseline z-score alone. **If the full four-model stack does not beat the simple z-score screen out of sample, use the simple screen.** Complexity must earn its place.
 10. **Paper-trade for two quarters before deploying capital.** Log every signal with a timestamp. Compare live signal quality to backtest expectation. Expect roughly 50–70% of backtest performance to survive contact with reality — if you get more, suspect a bug.
-

10. Layer 8 — Decision Surface (UI)

Six screens. The design principle: **the system ranks, the human decides — and the human can always see why.** Every score is expandable to its inputs. Nothing is a black box, because you cannot override what you cannot inspect.

1. Regime Dashboard (*home*) Regime label + probability gauge; recommended gross exposure vs. actual; the earnings-yield-minus-T-bill spread chart (the hero metric); macro variable heatmap with z-scores; ECT half-life; IRF panels; event calendar with historical CAR overlays.

2. Screener Sortable universe table: Composite, M1 α + t, M2 (with pillar breakdown on hover), M3 verdict, liquidity band, days-to-build. Filters for sector, tier, regime fit. Red-flag badges shown inline. Diff view: *what changed since yesterday, and why*.

3. Stock Deep Dive — four tabs, one per model

- *Carhart*: factor loadings bar chart, rolling α with confidence band, R^2 , residual vol, style fingerprint radar
- *Fundamentals*: four pillars, 5-year ratio history, peer comparison, SOTP waterfall for conglomerates, red-flag register
- *Timing*: price chart with 52-week high anchor, signal battery breakdown, branch label, entry band overlay
- *Macro fit*: sector sensitivity, regime-conditional expected return, relevant upcoming events

Plus a **thesis panel**: expected return, fair value, margin of safety, target size, days-to-build, exit triggers — all pre-populated, all editable.

4. Portfolio & Risk Positions with live P&L; **portfolio-level Carhart regression** (your true factor exposure vs. target); sector and liquidity concentration; contribution to risk by position; exit triggers approaching.

5. Backtest Lab Configure and run strategy variants; equity curve with per-regime shading; net vs. gross; DSR; capacity curve; parameter sensitivity heatmap.

6. Journal Every signal, every decision, every override — logged with the reasoning at the time. **This is the highest-value screen in the application over a three-year horizon.** It is the only mechanism that lets you distinguish “the model was wrong” from “I overrode a correct model,” and that distinction is what turns a tool into a skill.

11. Layer 9 — Monitoring, Decay & Kill Switches

Run nightly. Alert on breach.

Monitor	Metric	Alert threshold
Data integrity	Corporate action reconciliation mismatch	> 0.5% on any ticker
Data freshness	Hours since last successful ingest per source	> 36h

Signal decay	Rolling 12m rank IC of each model	Below 50% of backtest IC
Regime stability	Markov transition probability	Change > 0.25 in one month
Factor drift	Portfolio β vs. target	Any $ \Delta\beta > 0.4$
Live vs. backtest	Rolling 12m IR ratio	< 0.5 → review; < 0.3 → halt new entries
Crowding	Composite score correlation across held names	Pairwise mean > 0.7
Capacity	Position size vs. current ADV	Days-to-exit > 10

Model refresh cadence: factor returns daily · M1 regressions weekly · M2 on each earnings release + monthly re-rank · M3 daily · M4 monthly (or immediately on a CBSL policy decision or IMF milestone) · full re-estimation and validation quarterly.

12. Tech Stack & Build Sequence

Stack

Layer	Choice
Ingestion	Python + Apify actors (already operational)
Storage	PostgreSQL (+ TimescaleDB for price series)
Transform	dbt — testable, versioned, lineage-tracked
Econometrics	<code>statsmodels</code> (ARDL, VECM, VAR, Markov-switching), <code>arch</code> , <code>linearmodels</code> (Fama-MacBeth)
ML	<code>scikit-learn</code> (Elastic Net), <code>lightgbm</code> , <code>mlfinlab</code> -style purged CV
Orchestration	Prefect (or cron for v1 — do not over-engineer)
API	FastAPI
Frontend	React + Tailwind + Recharts
Cache	Redis for computed scores

Build sequence — ship a usable thing at every phase

Phase	Duration	Deliverable	Gate to proceed
1. Data spine	3–4 weeks	PIT store, corporate actions, universe gates, reconciliation tests	Reconciliation passes 30 consecutive days
2. Factors + M2	3–4 weeks	CSE factor library; M2 baseline z-score screen; simple ranked table UI	M2 baseline shows positive rank IC out of sample
3. M4 macro	3–4 weeks	ARDL/VECM, regime classifier, sector sensitivity, regime dashboard	Regime classifier correctly labels historical periods
4. M1 + M3	3–4 weeks	Dimson-corrected Carhart, momentum/contrarian battery, fusion logic	Full stack beats M2-alone out of sample
5. Execution + validation	3–4 weeks	Sizing, exits, backtest lab, journal, monitoring	Two quarters of paper trading

Ship Phase 2 and use it. A working value-quality screen with clean point-in-time data is already better than what most CSE participants have. Phases 3–5 add edge; Phase 1–2 is edge. The most common failure mode for a project like this is building all four models on dirty data and never shipping.

13. Failure-Mode Register

#	Failure mode	Why it's likely here	Mitigation
1	Look-ahead bias via restated financials	CSE restatements are common and quietly overwritten	Immutable versioned PIT store; <code>first_available_date</code> on every query
2	Downward-biased betas from non-trading	Structural feature of the CSE	Dimson lead/lag aggregation, mandatory
3	Fake value from revalued property	Endemic in hotels, plantations, property	"Hard book" B/M as primary

4	Backtest alpha destroyed by costs	~2.24% round-trip fees + spread + impact	Cost model in every backtest; 8% expected-return floor; turnover cap
5	Overfitting on small N	~130 names, ~12 usable years	Regularisation, purged walk-forward, Deflated Sharpe, baseline-must-be-beaten rule
6	Structural break contamination (2022 default)	Breaks every long-window estimate	Zivot-Andrews; break flags in UI; sub-period reporting
7	Momentum crash on the rebound	Classic, and violent in frontier markets	M4 regime gate on momentum weight
8	Signal double-counting across models	The models share underlying data	M1's redundancy check at Stage 4 of fusion
9	Owning what you cannot sell	Illiquidity is the moat <i>and</i> the trap	Liquidity cap in sizing; days-to-exit monitor
10	Related-party value extraction	Family-controlled conglomerate structures	Pillar D hard veto, not a scored input
11	Silent data source breakage	Scrapers break when sites change	Freshness monitors; reconciliation; quarantine on failure
12	Human overriding correct signals	The most common real-world failure	Journal every override; quarterly review of override P&L vs. model P&L

14. Where the Profitability Actually Comes From

A closing note, because the temptation with a four-model stack is to believe the sophistication is the edge. It isn't. Ranked by expected contribution to your net returns:

1. **Clean point-in-time data** (~30% of the edge). Almost nobody on the CSE has this. It is unglamorous, expensive in effort, and it is the moat.
2. **Cost discipline and the 8% expected-return floor** (~25%). Low turnover in a 2.24%-round-trip market is worth more than any model improvement.
3. **The macro regime layer** (~20%). On the CSE, getting exposure right beats getting

selection right, because the market moves as a bloc with the rate cycle.

4. **Earnings-integrity veto** (~15%). Avoiding the bad half of a family-controlled market beats picking the good half.
5. **Everything else — factor construction, momentum refinement, ML** (~10%).

The models are how you make the process **repeatable and measurable**, which is what lets you improve it. But the money is in the boring layers. Build them first.

Appendices to be populated during Phase 1: A — full data dictionary; B — formula reference; C — CSE sector classification mapping; D — parameter defaults and permitted ranges.